CS6450

AFL: A Single-Round Analytic Approach for Federated Learning with Pre-trained Models

(CVPR-2024)

Run He1, Kai Tong1, Di Fang1, Han Sun2,3, Ziqian Zeng1, Haoran Li4, Tianyi Chen5, Huiping Zhuang1∗ 1 South China University of Technology 2 Tsinghua University 3 Beijing National Research Center for Information Science and Technology 4 The Hong Kong University of Science and Technology 5 Microsoft

Course Instructor
Prof. Dr. C. Krishna Mohan

Presented by
Atish Kadam
CS25MTECH14003

Assigned TA
Ishu Priya

భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

Motivation

Federated Learning (e.g., FedAvg) faces four major **"bottlenecks"** :

1. **Data Heterogeneity (Non-IID):** Models struggle to converge when clients have very different data distributions.
2. **Large Client Count:** Performance often degrades when scaling to 1,000+ clients
3. **Slow Convergence:** It takes many rounds of communication to reach a stable state.
4. **High Communication Cost:** Sharing weights back and forth multiple times is bandwidth-intensive.

Problem Statement

- The research objective is to develop a novel FL framework that overcomes these limitations by providing analytical, closed-form solutions for efficient and robust training.

- AFL aims to address these limitations by leveraging analytic learning principles to develop a robust, efficient, and hyperparameter-free FL framework.

Challenges

1. Implementing Analytic learning at both client and server side

2. We must have pre-trained models for data training at client side.

3. Optimized Aggregation method is necessary at server side to aggregate in single round.

Related Work

1. Traditional Federated Learning (FL)

Existing FL Research designed to fix the "drift" that happens when clients have different data (non-IID).

- **Constraint-Based (FedProx, FedNova):** These methods add mathematical "leashes" to local training to prevent a client's model from wandering too far away from the global average.
- **Adaptive Weighting (FedLAW, FedCDA):** Instead of simple averaging, these methods try to learn the "importance" of each client. However, they often require a "proxy dataset" (a small slice of data kept on the server), which can be a privacy risk or hard to collect.
- **Structural Alignment (Fed2):** These focus on making sure the "neurons" or parameters across different clients actually represent the same features before they are averaged.
- **Distillation-Based:** These use **Knowledge Distillation**, where the server averages the *outputs* (logits) of client models rather than their raw weights.

➢ **The AFL Critique:** The argument is that all the above still rely on **gradients**, which means they still suffer from slow convergence, high communication costs (multiple rounds), and sensitivity to the number of clients.

Proposed Method

- We propose **Analytic Federated Learning (AFL)**, a gradient-free FL framework utilizing analytical closed-form solutions for both local client training and aggregation.

- AFL incorporates a **one-epoch client** training stage where a **pre-trained network's** classification head is reformulated into a localized linear regression problem, yielding an LS-based solution.

- The aggregation stage introduces an **Absolute Aggregation (AA) law,** derived analytically, enabling single-round aggregation without approximation or parameter tuning.

Analytic Learning (AL)

While standard neural networks use Gradient Descent (iteratively nudging weights to reduce error), Analytic Learning uses Linear Algebra to calculate the optimal weights in one or very few steps.

In traditional training, you update weights **w** slowly to minimize a loss function. In AL, we treat the neural network layers as a system of linear equations:

$$\boldsymbol{Xw} = \boldsymbol{Y}$$

Where **X** is your input data and **Y** is the target. To find w, instead of iterating, AL uses the **Moore-Penrose Pseudoinverse**

$$\boldsymbol{w} = \boldsymbol{X}^{\dagger}\boldsymbol{Y}$$

This is also known **as Least Squares (LS) estimation**. It finds the "best fit" weights mathematically rather than through repetitive training epochs.

AL solves several "headaches" common in standard AI training:

- **No Gradient Issues**
- **One-Epoch Training**
- **Stability**

Methodology – Analytical Federated Learning

1. Local Stage: Localized Analytic Learning

Step 1: Feature Extraction

$$x_{k,j} = f_{\text{backbone}}(X_{k,j}, \Theta) \qquad (1)$$ **Embedding Vector**

Step 2: Data Stacking

$$\mathbf{X}_k = \begin{bmatrix} \mathbf{x}_{k,1} \\ \vdots \\ \mathbf{x}_{k,N_k} \end{bmatrix}, \quad \mathbf{Y}_k = \begin{bmatrix} \text{onehot}(y_{k,1}) \\ \vdots \\ \text{onehot}(y_{k,N_k}) \end{bmatrix}$$ **(2) Embedding Matrix**

Step 3: Optimization via Analytic Learning (AL)

The goal is to find the weights W_k that minimize the Mean Square Error (MSE) loss:

$$L(W_k) = ||Y_k - X_k W_k||_F^2 \qquad (3)$$

The optimal weight estimation $\widehat{W_k}$ is found using the Moore-Penrose (MP) inverse (†):

$$\widehat{W_k} = \text{argmin}_{W_k} L(W_k) = X_k^{\dagger} Y_k \qquad (4)$$

Methodology – Analytical Federated Learning

2. Aggregation Stage: Absolute Aggregation

Theorem 1. Absolute Aggregation Law:

Client -1 U
Client -2 V

Assumption –
Xk – Input is Assumed to be Full-Column Rank.

The Aggregation Rule
The global weight is a linear combination of local weights:

$$\boldsymbol{\widehat{W}} = \boldsymbol{\mathcal{W}_u \widehat{W}_u} + \boldsymbol{\mathcal{W}_v \widehat{W}_v}$$

$\boldsymbol{\widehat{W}}$ **-** The **Global Model**
$\boldsymbol{\widehat{W}_u}$**-** The model trained locally by Client u ($\widehat{W}_u = X_u^{\dagger} Y_u$)
$\boldsymbol{\widehat{W}_v}$ **-** The model trained locally by Client v ($\widehat{W}_v = X_v^{\dagger} Y_v$)
$\boldsymbol{\mathcal{W}_u, \mathcal{W}_v}$ **-** The Aggregation Matrices

Methodology – Analytical Federated Learning

2. Aggregation Stage: Absolute Aggregation

The Weighting Matrices

These matrices $(\mathcal{W}_u, \mathcal{W}_v)$act as the "importance" or "contribution" factors for each client:

$$\mathcal{W}_u = I - R_u C_v - R_u C_v (C_u + C_v)^{-1} C_v$$
$$\mathcal{W}_v = I - R_v C_u - R_v C_u (C_u + C_v)^{-1} C_u$$

$\boldsymbol{I}$ **(Identity Matrix):** Acts like the number "1" in matrix algebra.

C_u, C_v**(Gram Matrices):** Computed as $X_u^\top X_u$ and $X_v^\top X_v$. These represent the feature correlation of the local datasets.

$\boldsymbol{R_u}, \boldsymbol{R_v}$ **(Inverses):** Computed as C_u^{-1} and C_v^{-1}.

Methodology – Analytical Federated Learning

2. Aggregation Stage: Absolute Aggregation

Scaling to Multiple Clients (K Clients)

The accumulated aggregation weight that has aggregated k clients –

$$\widehat{W_{agg,k}} = \mathcal{W}_{agg} \widehat{W_{agg,k-1}} + \mathcal{W}_{k} \widehat{W_{k}}$$

- To include client kk, combine the previous aggregate with the new client weight.
- Aggregation is **pairwise**, not global averaging.
- Same AA law used repeatedly.

Limitation

This may not hold in the large client number scenario where each client has limited data rendering the full-column rank assumption invalid.

Methodology

3. RI (Regularization Intermediary) Process: AA Law in Rank-deficient Scenario

To this end, we include an regularization term controlled by γ in the objective function, i.e.

$$\mathcal{L}(W_k^r) = ||Y_k - X_k W_k^r||_F^2 + \boldsymbol{\gamma}||\boldsymbol{W_k^r}||_{\boldsymbol{F}}^{\boldsymbol{2}} \quad \leftarrow \textbf{Extra term}$$

The Deviation: Regularized vs. True Solution $\widehat{\boldsymbol{W^r_{agg,k}}} \neq \widehat{\boldsymbol{W_{agg,k}}}$

Theorem 2. RI-AA Law: The relation between $\widehat{\boldsymbol{W^r_{agg,k}}}$ and $\widehat{\boldsymbol{W_{agg,k}}}$ follows:

$$\widehat{\boldsymbol{W_{agg,k}}} = \left(\boldsymbol{C^r_{agg,k}} - \boldsymbol{k\gamma I}\right)^{-1} \boldsymbol{C^r_{agg,k}} \widehat{\boldsymbol{W^r_{agg,k}}}$$

Architecture

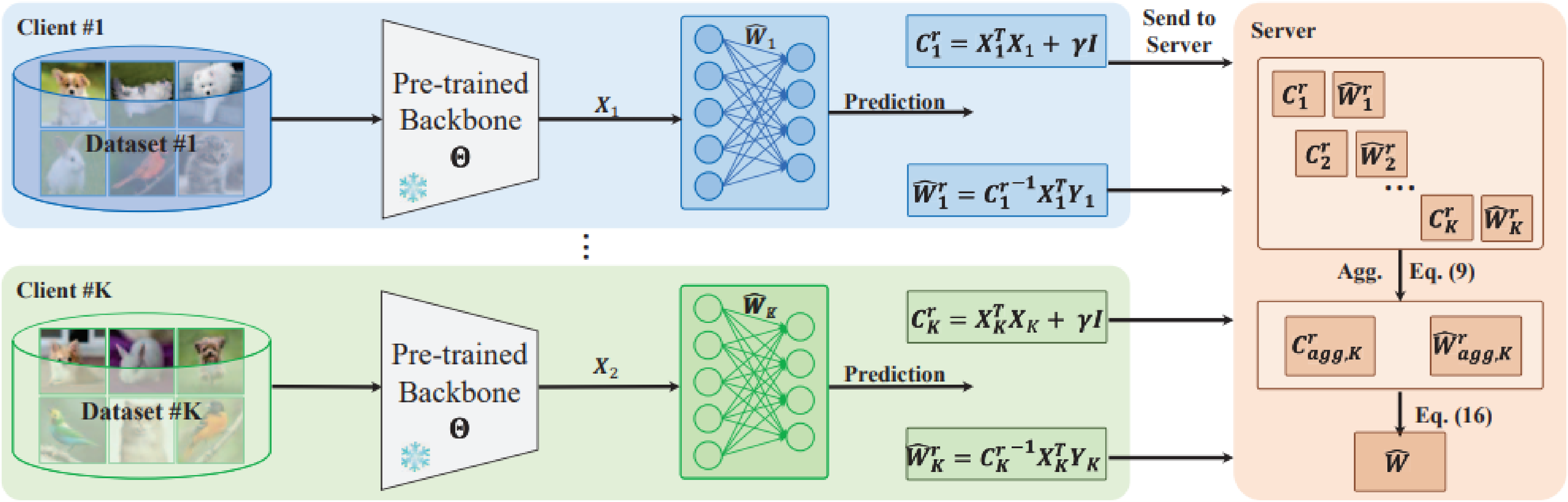

Figure 1. An overview of the AFL. During the local stage, each client calculates C_k^r and $\hat{W}_k^r$ based on the same pre-trained backbone and its own dataset. The server obtained the $C_{\text{agg},K}^r$ and $\hat{W}_{\text{agg},K}^r$ then get $\hat{W}$ in the aggregation stage.

Datasets Used

We validate the baselines and our proposed AFL in 3 popular benchmark datasets in FL:

- **CIFAR-10**
- **CIFAR-100**
- **Tiny-ImageNet.**

Results

Invariance to Heterogeneity

Dataset	Setting	FedAvg	FedProx	MOON	FedGen	FedDyn	FedNTD	FedDisco	AFL
CIFAR-10	$\alpha = 0.1$	$64.02_{\pm 0.18}$	$64.07_{\pm 0.08}$	$63.84_{\pm 0.03}$	$64.14_{\pm 0.24}$	$64.77_{\pm 0.11}$	$64.64_{\pm 0.02}$	$63.83_{\pm 0.08}$	$\mathbf{80.75}_{\pm 0.00}$
	$\alpha = 0.05$	$60.52_{\pm 0.39}$	$60.39_{\pm 0.09}$	$60.28_{\pm 0.17}$	$60.65_{\pm 0.19}$	$60.35_{\pm 0.54}$	$61.16_{\pm 0.33}$	$59.90_{\pm 0.05}$	$\mathbf{80.75}_{\pm 0.00}$
	$s = 4$	$68.47_{\pm 0.13}$	$68.46_{\pm 0.08}$	$68.47_{\pm 0.15}$	$68.24_{\pm 0.28}$	$73.50_{\pm 0.11}$	$70.24_{\pm 0.11}$	$65.04_{\pm 0.11}$	$\mathbf{80.75}_{\pm 0.00}$
	$s = 2$	$57.81_{\pm 0.03}$	$57.61_{\pm 0.12}$	$57.72_{\pm 0.15}$	$57.02_{\pm 0.18}$	$64.07_{\pm 0.09}$	$58.77_{\pm 0.18}$	$58.78_{\pm 0.02}$	$\mathbf{80.75}_{\pm 0.00}$
CIFAR-100	$\alpha = 0.1$	$56.62_{\pm 0.12}$	$56.45_{\pm 0.22}$	$56.58_{\pm 0.02}$	$56.48_{\pm 0.17}$	$57.55_{\pm 0.08}$	$56.60_{\pm 0.14}$	$55.79_{\pm 0.04}$	$\mathbf{58.56}_{\pm 0.00}$
	$\alpha = 0.01$	$32.99_{\pm 0.20}$	$33.37_{\pm 0.09}$	$33.34_{\pm 0.11}$	$33.09_{\pm 0.09}$	$36.12_{\pm 0.08}$	$32.59_{\pm 0.21}$	$25.72_{\pm 0.08}$	$\mathbf{58.56}_{\pm 0.00}$
	$s = 10$	$55.76_{\pm 0.13}$	$55.80_{\pm 0.16}$	$55.70_{\pm 0.25}$	$60.93_{\pm 0.17}$	$\mathbf{61.09}_{\pm 0.09}$	$54.69_{\pm 0.15}$	$54.65_{\pm 0.09}$	$58.56_{\pm 0.00}$
	$s = 5$	$48.33_{\pm 0.15}$	$48.29_{\pm 0.14}$	$48.34_{\pm 0.19}$	$48.12_{\pm 0.06}$	$\mathbf{59.34}_{\pm 0.11}$	$47.00_{\pm 0.19}$	$45.86_{\pm 0.18}$	$58.56_{\pm 0.00}$
Tiny-ImageNet	$\alpha = 0.1$	$46.04_{\pm 0.27}$	$46.47_{\pm 0.23}$	$46.21_{\pm 0.14}$	$46.27_{\pm 0.14}$	$47.72_{\pm 0.22}$	$46.17_{\pm 0.16}$	$47.48_{\pm 0.06}$	$\mathbf{54.67}_{\pm 0.00}$
	$\alpha = 0.01$	$32.63_{\pm 0.19}$	$32.26_{\pm 0.14}$	$32.38_{\pm 0.20}$	$32.33_{\pm 0.14}$	$35.19_{\pm 0.06}$	$31.86_{\pm 0.44}$	$27.15_{\pm 0.10}$	$\mathbf{54.67}_{\pm 0.00}$
	$s = 10$	$39.06_{\pm 0.26}$	$38.97_{\pm 0.23}$	$38.79_{\pm 0.14}$	$38.82_{\pm 0.16}$	$41.36_{\pm 0.06}$	$37.55_{\pm 0.09}$	$38.86_{\pm 0.12}$	$\mathbf{54.67}_{\pm 0.00}$
	$s = 5$	$29.66_{\pm 0.19}$	$29.17_{\pm 0.16}$	$29.24_{\pm 0.30}$	$29.37_{\pm 0.25}$	$35.18_{\pm 0.18}$	$29.01_{\pm 0.14}$	$27.72_{\pm 0.18}$	$\mathbf{54.67}_{\pm 0.00}$

Two challenging heterogeneity scenarios were tested:

LDA Partition (NIID-1)Data distribution controlled by Dirichlet parameter α

Smaller α → More severe heterogeneity

Sharding (NIID-2)Controlled by number of shards per client (s)

Smaller s → More severe heterogeneity

Results

Invariance to number of clients

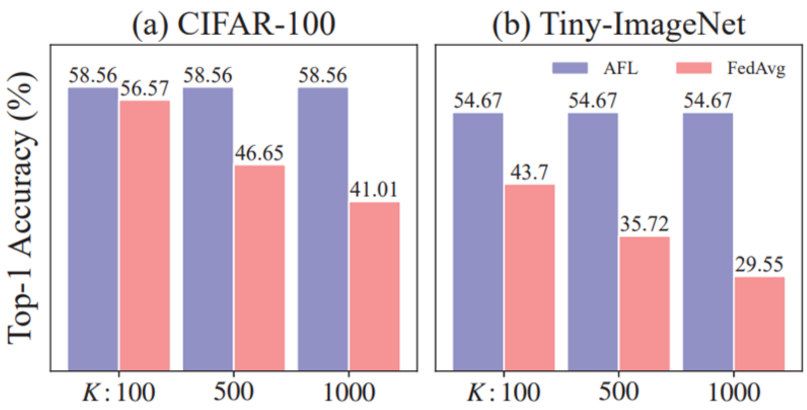

Figure 2. Accuracy over various number of clients.

Invariance to Heterogeneity

Table 2. The top-1 classification accuracy (%) of AFL and FedAvg under different data heterogeneity.

Acc. (%)	$\alpha = 0.005$	$\alpha = 0.01$	$\alpha = 0.1$	$\alpha = 1$	IID
FedAvg	24.74	33.09	56.57	57.72	57.89
AFL	**58.56**	**58.56**	**58.56**	**58.56**	**58.56**

Results

Fast Training with Single-round Aggregation

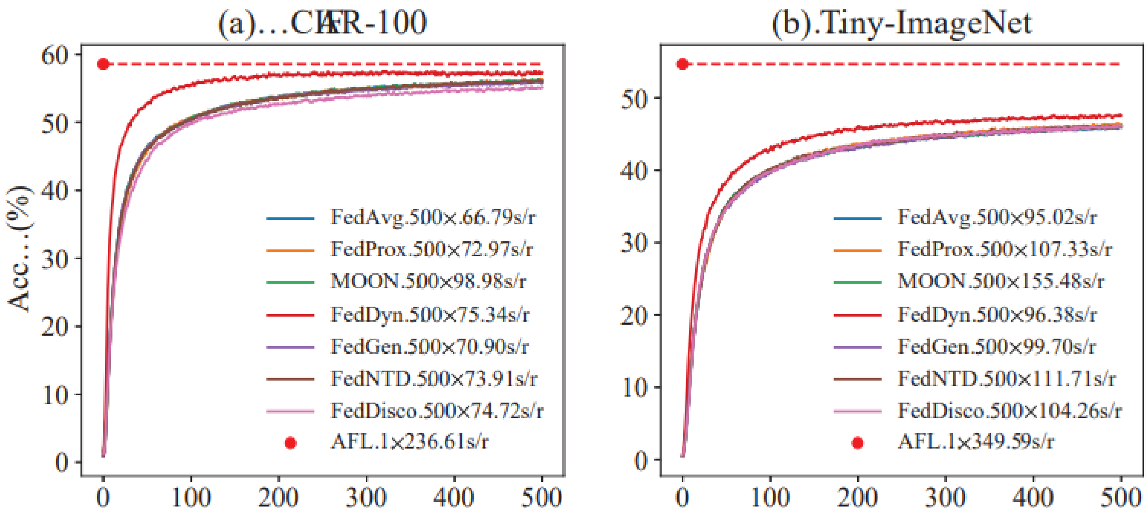

Figure 3. Accuracy curves with communication rounds. Average training time is reported in the legends.

Limitations and Future Work

1. **Partially Participating and Stragglers**
- When clients engage partially
- AFL needs to wait for all the clients
- Hampers the AFL's overall efficiency

2. **Linear Assumptions of AFL**
- The AFL is established upon linear classifiers
- May be less effective with nonlinear data distribution

Conclusion

Gradient-Free Efficiency: Utilizes analytical solutions to achieve one-epoch local training and single-round aggregation, drastically reducing communication overhead.

Theoretical Rigor: Backed by the AA Law for single-round convergence and the RI Process to ensure optimality in large-scale, rank-deficient scenarios.

Invariance Properties: Proven resistance to data heterogeneity (Non-IID) and client scaling, ensuring consistent performance regardless of how data is partitioned.

Final Result: A fast, robust, and mathematically grounded framework that outperforms traditional iterative FL methods in both speed and stability.

References

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Thank You!